

LCMSET

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1.1

$S \quad \text{LCM}(S) \quad S$.
 $T \quad \text{LCMSET}(T)$

$$\text{LCMSET}(T) = \{\text{LCM}(S) : \emptyset \neq S \subseteq T\}.$$

$A, B \quad \text{LCMSET}(A) = \text{LCMSET}(B)$.

1.2

- $|A|, |B| \leq 50$
- $\leq 10^9$
-

2 : Canonical Form

2.1 Join-Semilattice

$\text{LCMSET}(A) \quad \text{LCM}(=\text{join}) \quad \text{finite join-semilattice} \quad \text{join-semilattice} \quad , \quad \text{lattice}$
 $(\text{join-irreducible})$.

2.2 Redundancy

$a \in A \quad A \quad \text{redundant}$

$$\exists S \subseteq A \setminus \{a\}, S \neq \emptyset, \text{ s.t. } a = \text{LCM}(S).$$

$A' := \{a \in A : a \text{ is non-redundant in } A\} \quad , \quad A' \quad \text{LCMSET}(A) \quad \text{join-irreducible}$.

2.3

Theorem

$$\text{LCMSET}(A) = \text{LCMSET}(B) \iff A' = B'.$$

Proof Sketch

$L = \text{LCMSET}(A) \quad \text{join-irreducible} \quad J(L)$.

Step 1. $J(L) \subseteq A$. $c \in J(L) \quad c = \text{LCM}(S), S \subseteq A$ minimal $S \quad , \quad |S| \geq 2 \quad S = S_1 \sqcup S_2$

minimality $\text{LCM}(S_1), \text{LCM}(S_2) < c$ c join-irreducible . $|S| = 1, c \in A$.
Step 2. $A' = J(L)$.

- $a \in A' \Rightarrow a \in J(L)$: $a = c_1 \vee c_2, c_1, c_2 \in L, c_1, c_2 < a$. $c_i = \text{LCM}(T_i), T_i \subseteq A$.
 $c_i < a \quad a \notin T_i. \quad a = \text{LCM}(T_1 \cup T_2), T_1 \cup T_2 \subseteq A \setminus \{a\}$ — non-redundant .
- $a \notin A' \Rightarrow a \notin J(L)$: $a = \text{LCM}(S), S \subseteq A \setminus \{a\}, |S| \geq 2$ minimal. a join .

Step 3. $L \cong J(L)$ (finite lattice).

$$L_A = L_B \iff J(L_A) = J(L_B) \iff A' = B'. \quad \blacksquare$$

3 Redundancy Lemma

(2^n) lemma.

! Lemma

$a \in A$

$$D_a := \{x \in A \setminus \{a\} : x \mid a\}$$

.

$$a \text{ is redundant in } A \iff \text{LCM}(D_a) = a.$$

(, $D_a = \emptyset$ a non-redundant.)

💡 Proof

$(\Leftarrow) D_a \subseteq A \setminus \{a\} \quad \text{LCM}(D_a) = a$ redundant.

$(\Rightarrow) a = \text{LCM}(S), S \subseteq A \setminus \{a\}, S \neq \emptyset$.

- $x \in S \quad x \mid \text{LCM}(S) = a \quad x \in D_a. \quad S \subseteq D_a, \quad \text{LCM}(S) \mid \text{LCM}(D_a), \quad a \mid \text{LCM}(D_a).$
- $y \in D_a \quad y \mid a \quad \text{LCM}(D_a) \mid a.$
- $\text{LCM}(D_a) = a. \quad \blacksquare$

i

“ a LCM ” , $S \subseteq D_a$. , a LCM .

4

4.1 :

.

$$A = (a_1 < a_2 < \dots < a_n) \quad , \quad a_i$$

$$D_{a_i} \subseteq \{a_1, a_2, \dots, a_{i-1}\}$$

$$(a_i \quad a_i \quad).$$

Incremental Approach :

- 1.
2. LCM
3. $L = a_i$ redundant **early termination**

4.2 Early Termination

i Claim

LCM $L = a_i$ redundant .

💡 Proof

$L = \text{LCM}(\{x \in \{a_1, \dots, a_{i-1}\} : x \mid a_i, x \text{ processed so far}\}) = a_i$. $A \setminus \{a_i\}$
 $\text{LCM} = a_i$, redundant $(\exists S \subseteq A \setminus \{a_i\}, \text{LCM}(S) = a_i)$. $\blacksquare$

4.3

a_i :

- : $O(i)$
- LCM : $O(\log V)$, $V = \max$

: $O(n^2 \log V)$. $n \leq 50$, $V \leq 10^9$ $\sim 10^4$.

4.4

redundancy (deterministic) , A', B' $O(n)$ ().

5

5.1 C++

```

#include <stdio.h>

using ull = unsigned long long int;

ull _get_gcd(ull a, ull b)
{
    if(b == 0) return a;
    return _get_gcd(b, a % b);
}

int main(void)
{
    int t;
    scanf("%d", &t);
    while(t--)
    {
        int A[50], B[50];
        int a, b;
        scanf("%d%d", &a, &b);
        for(int i = 0; i < a; ++i) scanf("%d", &A[i]);
        for(int i = 0; i < b; ++i) scanf("%d", &B[i]);

        // (1)      (Insertion Sort)
        auto _insertion_sort = [](int n, int* arr) -> void {
            for(int j = 1; j < n; ++j)
            {
                auto key = arr[j];
                auto i = j - 1;
                while(i >= 0 && arr[i] > key)
                {
                    arr[i + 1] = arr[i];
                    --i;
                }
                arr[i + 1] = key;
            }
        };

        _insertion_sort(a, A);
        _insertion_sort(b, B);

        int _a, _b, _A[50], _B[50];
        _a = _b = 0;

        // (2) Redundant      canonical form
        auto _get_frozen_set = [](int n, int* src, int* dst) -> int {
            int cnt = 0;

```

```

    auto _get_lcm = [] (ull a, ull b) -> ull {
        return (a * b) / _get_gcd(a, b);
    };

    for(int i = 0; i < n; ++i)
    {
        ull L = 1;
        for(int j = 0; j < cnt; ++j)
        {
            if((src[i] % dst[j]) != 0) continue;
            L = _get_lcm(L, dst[j]);
            if(L == src[i])
            {
                break; // early termination: redundant
            }
        }
        if(L != src[i])
        {
            dst[cnt++] = src[i]; // non-redundant
        }
    }

    return cnt;
};

_a = _get_frozen_set(a, A, _A);
_b = _get_frozen_set(b, B, _B);

// (3) canonical form
if(_a != _b)
{
    printf("!\n");
}
else
{
    int i = 0;
    for(; i < _a; ++i)
    {
        if(_A[i] != _B[i]) break;
    }
    printf(i == _a ? "=\n" : "!\n");
}
}
}

```

5.2

5.2.1 Step 1:

Insertion Sort . $n \leq 50$ $O(n^2)$.

5.2.2 Step 2: Canonical Form

```
for(int i = 0; i < n; ++i)
{
    ull L = 1;
    for(int j = 0; j < cnt; ++j)
    {
        if((src[i] % dst[j]) != 0) continue; // skip
        L = _get_lcm(L, dst[j]); // LCM
        if(L == src[i]) break; // early termination
    }
    if(L != src[i]) dst[cnt++] = src[i]; // non-redundant
}
```

invariant:

- $dst[0..cnt-1]$ $a_1, \dots, a_{i-1}$ non-redundant .
- $src[i]$ LCM L .
- $L == src[i]$ redundant (Lemma).
- $L != src[i]$ non-redundant dst .

5.2.3 Step 3: Canonical Form

, .

5.3 ()

Test	A	B	A'	B'	
1	$\{2, 3, 4, 6\}$	$\{1, 2, 3, 4\}$	$\{2, 3, 4, 6\}?$	$\{1, 2, 3, 4\}?$	=
2	$\{10, 11\}$	$\{11\}$	$\{10, 11\}$	$\{11\}$	!
3	$\{4, 8, 16\}$	$\{2, 4, 8, 16\}$	$\{4, 8, 16\}$	$\{2, 4, 8, 16\}$	!

Test 1

- $A = \{2, 3, 4, 6\}$: 6 LCM(2, 3) = 6 redundant. $A' = \{2, 3, 4\}$. 4 $\{2, 3\}$ LCM(=6) non-redundant. 2 $\{3\}$, 3 $\{2\}$. $A' = \{2, 3, 4\}$.
- $B = \{1, 2, 3, 4\}$: 1 LCM (LCM). LCMSET(B) 1 $\{1\}$

$$\text{LCM}(\{1\}) = 1 \quad . \quad \text{LCMSET}(A) = 1 \quad .$$

$\text{LCMSET}(A) \neq \text{LCMSET}(B)$, $\quad = \quad . \quad .$
 $\rightarrow \quad = \quad . \quad \text{LCMSET} \quad . \quad \{1, 2, 3, 4\} \quad \text{LCMSET} \quad : \{1, 2, 3, 4, \text{LCM}(2, 3) =$
 $6, \text{LCM}(2, 4) = 4, \text{LCM}(3, 4) = 12, \dots\} = \{1, 2, 3, 4, 6, 12\}. \quad \{2, 3, 4, 6\} \quad \text{LCMSET} =$
 $\{2, 3, 4, 6, 12\}. \quad ! \quad .$
 $: \quad = \quad . \quad . \quad \text{Accept} \quad .$

6

6.1

- **Incremental Approach:** canonical form
- **Cut-and-Paste Lemma** : redundancy $\rightarrow$

6.2

- **Join-semilattice:** LCM
- **Join-irreducible elements:** lattice
- **Canonical form:**

6.3 /

-
- LCM + early termination
- Insertion sort (n)

6.4

	$O(n^2)$
Canonical form	$O(n^2 \log V)$
	$O(n)$
(per test case)	$O(n^2 \log V)$

7 — n

$n \leq 50 \quad O(n^2 \log V) \quad , \quad n \quad .$

7.1

vector :

$$a_i = \prod_p p^{e_{i,p}}$$

LCM max exponent:

$$\text{LCM}(S) = \prod_p p^{\max_{i \in S} e_{i,p}}$$

a_i redundant $p \mid a_i$, a_i $e_{j,p} = e_{i,p}$.

$O(n \cdot \pi(V) + n \cdot \sqrt{V})$, .

8

Canonical form	LCMSET canonical form non-redundant = join-irreducible elements $D_a (\quad)$ LCM + LCM + early termination $\rightarrow$ $O(n^2 \log V)$ per test case
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